WORKED SOLUTION

STEP II 1990, Question 10

Pure • Matrix Polynomial Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q10

The key idea

Determinants turn a matrix product into a product of ordinary numbers, and a product of real numbers is zero only if one of the factors is. For the cubic, $\mathbf{M}$ commutes with $\mathbf{I}$, so a matrix polynomial in $\mathbf{M}$ factorises exactly like the polynomial in x. Upper triangular matrices then behave very well: the diagonal entries of a product are just the products of the diagonal entries, so the diagonal is quick to deal with and only the top-right corner needs real work.

Solution

Matrices with $\mathbf{AB} = \mathbf{0}$

Since the determinant of a product is the product of the determinants,

$$\det \mathbf{A} \det \mathbf{B} = \det(\mathbf{AB}) = \det \mathbf{0} = 0.$$

The left-hand side is a product of two real numbers, so at least one of them is zero: $\det \mathbf{A} = 0$, or $\det \mathbf{B} = 0$, or both. That is the required result.

If $\det \mathbf{B} \neq 0$ then $\mathbf{B}^{-1}$ exists, and multiplying $\mathbf{AB} = \mathbf{0}$ on the right by $\mathbf{B}^{-1}$ gives

$$\mathbf{A} = \mathbf{ABB}^{-1} = \mathbf{0}\,\mathbf{B}^{-1} = \mathbf{0}.$$

So $\mathbf{A}$ must be the zero matrix.

Both determinants being zero is not enough, though. Take

$$\mathbf{A} = \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $\det \mathbf{A} = \det \mathbf{B} = 0$, but $\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq \mathbf{0}$.

Factorising the cubic

First factorise the polynomial $f(x) = x^3 + 2x^2 - 5x - 6$. Trying small divisors of 6:

$$\begin{aligned} f(-1) &= -1 + 2 + 5 - 6 = 0, \\ f(2) &= 8 + 8 - 10 - 6 = 0, \\ f(-3) &= -27 + 18 + 15 - 6 = 0, \end{aligned}$$

so $f(x) = (x+1)(x-2)(x+3)$, since the leading coefficient is 1.

Now expand $(\mathbf{M} + p\mathbf{I})(\mathbf{M} + q\mathbf{I})(\mathbf{M} + r\mathbf{I})$. Because $\mathbf{I}$ commutes with $\mathbf{M}$ (and $\mathbf{I}^2 = \mathbf{I}$), the expansion is exactly the same as for numbers:

$$\mathbf{M}^3 + (p+q+r)\mathbf{M}^2 + (pq+qr+rp)\mathbf{M} + pqr\,\mathbf{I}.$$

Matching with the polynomial above, we can take

$$p = 1, \qquad q = -2, \qquad r = 3$$

(in any order), and then $\mathbf{M}^3 + 2\mathbf{M}^2 - 5\mathbf{M} - 6\mathbf{I} = (\mathbf{M} + \mathbf{I})(\mathbf{M} - 2\mathbf{I})(\mathbf{M} + 3\mathbf{I})$ for every square matrix $\mathbf{M}$.

> **Ask yourself** Can we now say that one of the three factors must be the zero matrix?

No. The first part shows that a product of matrices can vanish without any factor vanishing; all we learn is that at least one factor is singular. So we have to compute.

The upper triangular solutions

Let $\mathbf{M} = \begin{pmatrix} a & c \\ 0 & b \end{pmatrix}$. Each factor is upper triangular with the same top-right entry c:

$$\mathbf{M} + k\mathbf{I} = \begin{pmatrix} a+k & c \\ 0 & b+k \end{pmatrix}.$$

For three upper triangular matrices with diagonals (d_i, e_i) and top-right entries all equal to c, multiplying out gives

$$\begin{aligned} &\begin{pmatrix} d_1 & c \\ 0 & e_1 \end{pmatrix}\begin{pmatrix} d_2 & c \\ 0 & e_2 \end{pmatrix}\begin{pmatrix} d_3 & c \\ 0 & e_3 \end{pmatrix} \\ &\quad = \begin{pmatrix} d_1d_2d_3 & c\,(d_1d_2 + d_1e_3 + e_2e_3) \\ 0 & e_1e_2e_3 \end{pmatrix}. \end{aligned}$$

With $d_1, d_2, d_3 = a+1, a-2, a+3$ and $e_1, e_2, e_3 = b+1, b-2, b+3$, the equation becomes three conditions:

$$\begin{aligned} (a+1)(a-2)(a+3) &= 0, \\ (b+1)(b-2)(b+3) &= 0, \\ c\,[(a+1)(a-2) + (a+1)(b+3) + (b-2)(b+3)] &= 0. \end{aligned}$$

The first two say $a, b \in \{-1, 2, -3\}$. Expanding the bracket in the third,

$$(a^2 - a - 2) + (ab + 3a + b + 3) + (b^2 + b - 6) = a^2 + ab + b^2 + 2(a + b) - 5.$$

Ask yourself Is there a smarter way to see when this bracket is zero than trying all nine pairs?

Yes. If $a \neq b$, the bracket is

$$\frac{a^3 - b^3}{a - b} + 2\frac{a^2 - b^2}{a - b} - 5\frac{a - b}{a - b} = \frac{f(a) - f(b)}{a - b},$$

which is 0 because $f(a) = f(b) = 0$. So when $a \neq b$ the third condition holds whatever c is.

If $a = b$, the bracket is $3a^2 + 4a - 5 = f'(a)$. This is -6, 15 and 10 at $a = -1, 2, -3$ respectively (none is zero, since f has three distinct roots), so the third condition forces $c = 0$.

So the solutions are:

- $\mathbf{M} = \begin{pmatrix} a & c \\ 0 & b \end{pmatrix}$ with a and b *different* numbers from $\{-1, 2, -3\}$ and c *any* real number (six families);
- the three scalar matrices $-\mathbf{I}$, $2\mathbf{I}$ and $-3\mathbf{I}$ (when $a = b$, which forces $c = 0$).

Checks

Take $a = -1$, $b = 2$, $c = 5$. Then $\mathbf{M} + \mathbf{I} = \begin{pmatrix} 0 & 5 \\ 0 & 3 \end{pmatrix}$, $\mathbf{M} - 2\mathbf{I} = \begin{pmatrix} -3 & 5 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{M} + 3\mathbf{I} = \begin{pmatrix} 2 & 5 \\ 0 & 5 \end{pmatrix}$. The product of the first two is $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ already, so the whole product vanishes for any c. On the other hand $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ gives $\mathbf{M}^3 + 2\mathbf{M}^2 - 5\mathbf{M} - 6\mathbf{I} = \begin{pmatrix} 0 & 15 \\ 0 & 0 \end{pmatrix} \neq \mathbf{0}$, matching $f'(2) = 15$. A computer search over many rational a, b, c found exactly the solutions above, and the factorisation was checked on random 3×3 integer matrices.

Common slips

- Concluding from $(\mathbf{M} + \mathbf{I})(\mathbf{M} - 2\mathbf{I})(\mathbf{M} + 3\mathbf{I}) = \mathbf{0}$ that $\mathbf{M}$ must be $-\mathbf{I}$, $2\mathbf{I}$ or $-3\mathbf{I}$. The first part is there to warn you against exactly this.
- Forgetting the case $a = b$, or assuming c is free there too.
- Writing the example for the first part with $\mathbf{AB} = \mathbf{0}$ by accident; check that your product really is non-zero.

Beyond STEP

The answer is a first look at the *minimal polynomial*: a matrix satisfies a polynomial with distinct roots exactly when it is diagonalisable with eigenvalues among those roots. Our $a \neq b$ solutions are diagonalisable for every c, while $\begin{pmatrix} a & c \\ 0 & a \end{pmatrix}$ with $c \neq 0$ is not, which is why it fails. The divided difference $\big(f(a) - f(b)\big)/(a - b)$ in the corner, becoming $f'(a)$ when $a = b$, is how functions of matrices are defined in general. Strang's *Introduction to Linear Algebra* covers eigenvalues and diagonalisation clearly.

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